Stat Methods Appl (2012) 21:485–516
DOI 10.1007/s10260-012-0193-4

Assessing the effect of the amount of financial aids to Piedmont firms using the generalized propensity score

Michela Bia · Alessandra Mattei

Accepted: 4 April 2012 / Published online: 1 May 2012
© Springer-Verlag 2012

Abstract Regional and national development policies play an important role to support local enterprises in Italy. The amount of financial aid may be a key feature for firms' employment policies. We study the impact on employment of the amount of financial aid attributed to enterprises located in Piedmont, a region in northern Italy, analysing small-sized firms and medium- or large-sized firms separately. We apply generalized propensity score methods under the unconfoundedness assumption that adjusting for differences in a set of observed pre-treatment variables removes all biases in comparisons by different amounts of financial aid. We find that the estimated effects are increasing with amount of financial aid for both small-sized and medium- or large-sized firms, whereas the marginal effects of additional incentives are decreasing with amount of financial aid for small-sized firms, and have an inverse J-shape for medium- or large-sized firms.

Keywords Continuous treatment · Dose-response function · Financialaid to firms · Generalized propensity score · Non-experimental methods · Policy evaluation

1 Introduction

Regional and national development policies are an important tool to support local enterprises in Italy. Active labour policies are generally designed to improve employment for disadvantaged groups (e.g., youths and women), who may have difficulties

M. Bia (✉)
CEPS/INSTEAD, 44 rue Emile Mark, 4620 Differdange, Luxembourg
e-mail: michela.bia@ceps.lu

A. Mattei
Department of Statistics, University of Florence, Viale Morgagni, 59, Florence, Italy

entering the labour market. Another class of interventions, often considered by policy makers, aims at removing barriers (such as credit rationing), which might limit pro-competitive development, investment, applied research for environment safeguard, commercial promotion of companies.

The analysis of public policies is to be set into a sequence of evaluation processes, including the definition of interventions (process analysis), the evaluation of implementation (performance analysis), and the evaluation of the policy effects (impact analysis). This study focuses on the last aspect, analysing the impact of financial aids allocated between 2001 and 2003 to Piedmont enterprises.[1] The economic measures we consider are grants or soft-loans and are based on regional, national, and EU co-financed funds. The availability of several types of policy measures through which financial aid is provided leads to a high degree of heterogeneity in contribution levels received by firms.

Most of the recent policy impact evaluation literature has focused on evaluating causal effects of a binary treatment (e.g., to be exposed or not to a policy). A weakness of this approach is that it would classify all treated firms (beneficiaries) in the same way, irrespective of the amount of financial aid they receive. As the amount of aid is related to the local labour market conditions or to firms' performances, the expectation is that firms receiving different amounts of contribution will differ in their labour market outcomes. For this reason, we argue that it is important to go beyond estimation of the causal effects of public policies employing a binary discrete intervention. In fact, estimating dose-response and treatment effect functions could provide more information regarding the effectiveness of the policies. For instance, this approach may help to uncover heterogeneities along different amounts of financial aids and provide some information on the optimal level of incentives.

Bondonio (2008) deals with these issues assigning a financial aid equal to zero to non-subsided firms, and then classifying the sample of firms according to the received incentive level through a categorical treatment variable. Then, he focuses on pair-wise comparisons of the effect of receiving a specific incentive level versus not receiving contributions. He finds that (i) there exists a positive impact on employment of receiving versus not receiving contributions, and (ii) that this causal effect increases with the incentive class a firm belongs to. Pair-wise comparisons may be potentially affected by the way the continuous treatment variable is discretized, restricting the set of marginal effects that can be evaluated. A distinctive feature of the present paper is, instead, that it assesses the impact of the intensity of economic supports, by using the amount of policy exposure as a continuous variable. Therefore, our causal estimands are quite different: we focus on dose-response functions and marginal effect functions.

Estimating a "dose-response function" of financial aids could provide more information regarding the effectiveness of the policies by uncovering heterogeneities in the contribution levels. We accomplish this by applying recent results for the analysis of continuous treatments, and highlighting the type of insights that can be learned about the effects of public policies when analyzed in their continuous scale.

[1] 'Piedmont' is a region of northern Italy with Turin as its capital.

Our treatment variable is the intensity of the incentive in Euro. The outcome variable we focus on is the employment gain score defined as the difference between post and pre-intervention employment level. Employment is generally considered a good proxy of social-economic wellbeing, and the measures we focus on are primarily geared toward improving local socio-economic development, in order to come up to social expectations. We use a difference-in-difference specification of the outcome in order to remove any potential bias coming from time-invariant unobserved factors correlated to the outcome and to the treatment variable (e.g., Heckman et al. 1998; Abadie 2005).

One difficulty with this analysis is that the amount of aid is not exogenous to the firms' characteristics, implying that firms exposed to different levels of the treatment variable can systematically differ in important ways other than the observed treatment. In order to adjust for systematic differences in background characteristics occurring between groups with different levels of the treatment variable, a key identifying assumption is that selection into levels of the treatment is random conditional on a set of observable pre-treatment variables (unconfoundedness). A commonly used approach for estimation of causal effects under unconfoundedness in a binary treatment setting is the use of the propensity score (Rosenbaum and Rubin 1983a). Over the last years, several extensions of propensity score-based methods have been developed. Imbens (2000) and Lechner (2001) extended propensity score matching to handle a categorical treatment variable. For an ordinal treatment variable, Joffe and Rosenbaum (1999) and Lu et al. (2001) developed and applied a method based on a scalar balancing score. Hirano and Imbens (2004) introduced the concept of generalized propensity score (GPS) and extended the results in Rosenbaum and Rubin (1983a) to the case of a continuous treatment. Imai and Van Dyk (2004) introduced a concept similar to the GPS, the "propensity function", by allowing for arbitrary treatment regimes. In this paper, we estimate average treatment effects (on the treated) of different contribution levels by using the GPS approach introduced by Hirano and Imbens (2004). More specifically, we first estimate the GPS using a generalized linear model, and then we estimate the dose response function using a flexible parametric form for the regression function of the outcome on the treatment and the GPS.

The data we use consist of an integrated data base, linking administrative archives (INPS; ISTAT—ASIA, and data from Service and Industry Census 2001, co-ordinated by ISTAT) with information concerning different kinds of intervention and the amount of financial aid received by Piedmont firms in the years 2001, 2002 and 2003.[2] These data contain information on the type of incentives allocated through alternative legislative measures as well as on the amount of financial aid received by each firm. In addition, the data contain information about participants, such as the province where the firm is located, the sector of activity, the handicraft status, the number of employees, the uni-located status, and the firm's age. The use of all these covariates strengthens the plausibility of the unconfoundedness assumption. Finally, these data provide an opportunity to estimate dose-response functions separately for small-sized and medium- or large-sized enterprises, in order to learn if there exist differences in

[2] The authors thank Daniele Bondonio for providing the data.

the returns to amount of financial aids between these two groups. This analysis may inform policy makers about how to improve the efficacy of financial aids to firms for different types of firms.

Our study contributes to the existing literature in several ways. First, we advance the literature on the evaluation of public interventions for firms in Italy (e.g., Battistin et al. 2001; Mealli and Pagni 2001; Meini 2001; Pellegrini 2001; Pellegrini and Carlucci 2003; Bondonio 2002; Bondonio and Greenbaum 2006; Bronzini et al. 2008), and the evaluation of aids to Piedmont firms (e.g., Bondonio 2008). Second, we join a number of authors who have investigated the problem of estimating causal effects from multi-valued or continuous treatment (e.g., Hirano and Imbens 2004; Kluve et al. 2007; Mitnik 2008; Bia and Mattei 2008; Adorno et al. 2007 and 2008; Flores et al. 2011). Finally, in analyzing small-sized and medium- or large-sized firms separately, we contribute to the literature on the heterogeneity of active policies. (e.g., Ichino et al. 2005; Flores et al. 2011).

The paper is organized as follows. Section 2 describes the data used for the analysis and the sources of financial aids provided to Piedmont firms in years 2001, 2002 and 2003, and presents some descriptive statistics. Section 3 describes the methodological approach we apply to estimate the causal effects of interest, and Sect. 4 discusses the results of the applied estimation strategy. Section 5 draws some conclusions.

2 Piedmont firms data

The data we use consist of an integrated data base, linking administrative archives (INPS; ISTAT—ASIA, and data from Service and Industry Census 2001, co-ordinated by ISTAT) with information concerning different kinds of intervention and the amount of financial aid received by Piedmont firms in the years 2001, 2002 and 2003 (see Bondonio 2008 for details on the data).

Our analyses are based on a sample of Piedmont firms which were active in 2003 and either (i) had fewer than 50 employees (small-sized firms) and received a financial aid ranging between 500 and 200,000 Euro; or (ii) had more than (or equal to) 50 employees (medium- or large-sized firms), worked on manufacturing business areas as non-handicraft enterprises, and received a financial aid ranging between 500 and 700,000 Euro. From this sample of firms we discarded 28 firms which experienced a high employment change between 2000 and 2003, either losing more than 75 employees, or recruiting more than 250 new employees. This sample selection focuses attention on the core group of firms reflecting the Italian business market, and excludes a small number of cases where there is ground for concern about the accuracy of the number of employees.

2.1 Grants and below-market-rate loans to the Piedmont firms

Economic supports provided to Piedmont firms between 2001 and 2003 include regional, adopted by regions, national and EU co-financed aids (Bondonio 2008). More specifically, data on aids to Piedmont firms include enterprises which received either automatic incentives or grants based on evaluation procedures (by announcement

or by call), through either grants or below-market-rate loans. Grants include investment tax relief grants (Law 266/97 (art. 14), Law 341/95 and Law 297/99), subsidies to R&D (Law 140/97), and capital or interest grants (Laws 488/92, 215/92, 236/93, 662/96, 226/99, 228/97). Investment tax relief grants are intended to stimulate private investment in research and business development, and to support disadvantaged areas by providing tax incentives. The law 140/97 is specifically designed to finance facilities for research and pro-competitive development. Capital or interest grants aim at promoting private accumulation in the less developed areas of Italy (Law 488/92), and providing economic support to firms located in risk areas (Laws 226/99, 228/97). The law 662/96 concerns the so-called 'Area Agreements', which are intended to encourage local business through agreements between public and private institutions, and between local and external actors.

Soft-loans on investments, employment and economic and financial performances are also provided by Laws 1329/65, 56/86, 67/94, 598/94, 95/95, 22/97, 266/97 (art. 8), and Regional Laws 67/94, 24/97, 28/93, 56/86. Finally, 'Docup Objective 2' measures are designed to support depressed area with high persistent unemployment and declining industrial production, by providing business investment incentives. More details on each specific legislative measure can be found in "Appendix".

In order to make grants and below-market-rate loans comparable, the second ones are turned into gross subsidy equivalent (GSE). Formally, let A be the total amount financed, then the actual financial aid is

$$T = \sum_{t=1}^{p} \frac{r_m - r_s}{(1 + r_m)^t} A + \sum_{t=p+1}^{M} \frac{f_m - f_s}{(1 + r_m)^t} \tag{1}$$

where p is the pre-depreciation period, M is the financing spell-length, r_m and r_s are the market and subsidized rates, f_m is the financing rent with market rate, and f_s is the financing depreciation rent with subsidized rate. GSE allows us to focus on the net benefit of the incentive.

Table 1 presents some summary statistics of the amount of financial aid for our sample of 4785 firms classified by size and type of measure (see Sect. 2 for details on the sample selection). On the whole, these measures provide about 230.6 million Euro. More than 65 % of these incentives are supplied by the Laws 140/97 (17.2 %), 266/97 art. 8 (16.0 %), 341/95 (13.8 %), 598/94 (12.4 %), and 1329/65 (9.5 %), and no sample firm receives a financial aid from the Law 95/95, aiming at promoting 'junior' entrepreneurs. On average, each small-sized firm receives about 70, 508 Euro from National measures, 31, 297 Euro from funds given to the Piedmont Region, 14, 880 Euro from regional measures and 20, 543 Euro from DocUP measures. Medium- or large-sized firms receive much higher incentives: on average each medium- or large-sized firm receives about 185, 211 Euro from National measures, 90, 145 Euro from funds given to the Piedmont Region, 29, 151 Euro from regional measures and 47, 566 Euro from DocUP measures. Enterprises may apply to more calls and so receive more than one contribution. However, in our sample, more than 60 % of the participants benefits of only one financial measure.

Table 1 Financial aids to Piedmont firms: Summary statistics

Policy measure	All firms		Small-sized firms		Medium- or Large-sized firms	
	N	*Mean*	*N*	*Mean*	*N*	*Mean*
Financial aid	4802	48242	4114	35416	688	124942
National policy measures	*357*	*102638*	*257*	*70508*	*100*	*185211*
L. 228/97 (art. 4)	7	98801	5	119296	2	47564
L. 226/99 (art. 3)	68	52475	54	38462	14	106524
L. 662/96	77	132703	48	89143	29	204803
L. 95/95	0		0		0	–
L. 236/93	7	14806	3	6487	4	21046
L. 488/92 (Industry)	141	131558	95	88182	46	221140
L. 297/99 (D.M. 593/00)	36	42112	22	35331	14	52767
L. 215/92	33	60434	33	60434	0	–
Policy measures adopted by regions	*3928*	*40885*	*3288*	*31297*	*640*	*90145*
L. 598/94 (art. 11)	653	43911	523	34048	130	83592
L. 1329/65	821	26918	663	20121	158	55441
L. 341/95	1244	25695	1035	21180	209	48054
L. 140/97	1283	31435	881	23966	402	47804
L. 266/97 (art. 14)	18	32893	18	32893	0	–
L. 266/97 (art. 8)	1448	25508	1238	22722	210	41932
Regional policy measures	*463*	*16082*	*424*	*14880*	*39*	*29151*
R.L. 67/94	5	5424	5	5424	0	–
R.L. 24/97 (art. 6)	156	38749	123	40042	33	33931
R.L. 22/97	109	8623	109	8623	0	–
R.L. 56/86	195	2226	189	2206	6	2860
DOCUP 2000-06 Ob. 2 areas	*1075*	*25093*	*894*	*20543*	*181*	*47566*
Measure 1.2a	473	33663	372	29320	101	49660
Measure 2.1b	29	121608	18	80115	11	189506
Measure 2.1d	428	10439	357	9710	71	14102
Measure 2.2b	127	17944	110	17263	17	22352
Measure 2.2c	208	3056	181	3069	27	2970
Measure 2.4c	25	4800	16	4585	9	5183
Measure 4.1b	6	3850	5	4379	1	1207

2.2 Descriptive analyses

Tables 2 and 3 present some summary statistics of the relevant variables for the sample of 4,114 small-sized firms and 688 medium- or large-sized firms, respectively, grouped by amount of financial aid. We analyse small-sized and medium- or large-sized firms separately because they appear to be highly heterogeneous with respect to both the

Table 2 Small-sized Piedmont firms: Means (Standard Errors in parenthesis)

Number of firms	Financial Aid Interval per 1000							
	4114	[0.5; 4) 459	[4; 5.5) 205	[5.5; 7) 229	[7; 20) 1195	[20; 50) 1045	[50; 100) 626	[100; 200) 355
Treatment variable								
Financial aid (Euro)	35416	2118	4819	6210	12658	32302	70270	139284
	(40144)	(1036)	(458)	(456)	(3744)	(8403)	(14621)	(28514)
Outcome variable								
Employment gain score	1.51	0.90	0.60	1.32	0.87	1.41	2.09	4.36
	(7.96)	(4.48)	(4.25)	(12.70)	(6.28)	(7.10)	(9.51)	(11.94)
Background variables								
Province								
Turin	0.53	0.37	0.56	0.52	0.52	0.55	0.57	0.57
Vercelli	0.04	0.04	0.02	0.03	0.05	0.03	0.04	0.04
Novara	0.09	0.14	0.07	0.05	0.08	0.1	0.07	0.06
Cuneo	0.11	0.13	0.13	0.12	0.11	0.09	0.09	0.12
Asti	0.05	0.07	0.03	0.06	0.05	0.05	0.04	0.04
Alessandria	0.09	0.15	0.11	0.09	0.08	0.07	0.1	0.1
Biella	0.05	0.04	0.03	0.04	0.05	0.06	0.06	0.04
Verbano-Cusio-Ossola	0.05	0.05	0.04	0.08	0.06	0.04	0.02	0.04
Sector of activity								
Manufacturing industry	0.98	0.99	0.98	0.97	0.98	0.98	0.98	0.98
No handicraft firm	0.59	0.41	0.45	0.45	0.54	0.63	0.75	0.80
Number of employees	14.74	9.89	10.49	11.12	12.52	16.22	18.79	21.73
Micro enterprise								
(<10 employees)	0.42	0.61	0.51	0.53	0.50	0.34	0.29	0.22
Uni-located firm	0.85	0.88	0.86	0.90	0.86	0.84	0.81	0.79
Birth year: after 2000	0.11	0.15	0.13	0.11	0.12	0.11	0.09	0.11

amount of the financial aid received (the treatment variable) as well as background characteristics, including employment gain score (the outcome variable), defined as the difference between employment level in 2003 and employment level in 2000.

Consistently with the analyses shown in previous section, Tables 2 and 3 suggest that the amount of financial aid heavily depends on firm size: small-sized enterprises receive on average a much lower incentive than the medium- or large-sized ones. For instance, more than 75 % of the small-sized firms receive an incentive lower than 50,000 Euro, and about 50 % of the medium- or large-sized firms receive a financial aid greater than 75,000 Euro. Moreover, financial aids seem to have a higher variability among medium- or large-sized enterprises. Despite receiving higher levels of financial aid on average, medium- or large-sized firms seem to experience a lower employment gain score than small-sized firms. This suggest that small-sized firms have a greater propensity to invest in employment.

Table 3 Medium- or large-sized Piedmont firms: Means (Standard Errors in parenthesis)

Number of firms	Financial Aid Interval per 1000					
	688	[0.5; 15) 77	[15; 35) 114	[35; 75) 150	[75; 200) 204	[200; 700) 143
Treatment variable						
Financial aid (Euro)	124942	9202	24852	54945	125532	339638
	(132238)	(4105)	(5707)	(11979)	(35822)	(125221)
Outcome variable						
Employment gain score	0.91	1.72	−5.92	−0.23	1.05	6.93
	(26.01)	(27.23)	(21.61)	(18.80)	(28.87)	(29.45)
Background variables						
Province						
Turin	0.52	0.31	0.46	0.49	0.53	0.70
Vercelli	0.04	0.05	0.05	0.03	0.04	0.04
Novara	0.11	0.21	0.14	0.15	0.09	0.01
Cuneo	0.11	0.13	0.11	0.11	0.13	0.06
Asti	0.03	0.05	0.02	0.03	0.02	0.04
Alessandria	0.07	0.09	0.07	0.05	0.06	0.06
Biella	0.10	0.09	0.11	0.11	0.10	0.08
Verbano-Cusio-Ossola	0.02	0.06	0.04	0.02	0.02	0.01
Number of employees	116.07	97.38	105.14	106.4	132.00	122.25
Medium-sized enterprise						
(50–250 employees)	0.94	0.96	0.96	0.93	0.90	0.96
Uni-located firm	0.58	0.56	0.62	0.61	0.55	0.57
Birth year: after 2000	0.03	0.05	0.03	0.04	0.03	0.01

Tables 2 and 3 also provide some information on firms' background characteristics. Small-sized firms have on average 14.74 employees and about 42 % of them are micro-enterprises with less than 10 employees. Most of these small-sized firms are located in Turin, are involved in manufacturing activities as non-handicraft enterprises, and are uni-located in Piedmont. In the sample, young firms are only few: only the 11 % of the small enterprises were born after 2000. Medium- or large-sized firms have on average 116.07 employees, and most of them are medium enterprises with less than 250 employees. More than 50 % of these medium- or large-sized firms are located in Turin and are uni-located in Piedmont. As we could expect, the proportion of young firms is particularly low among medium- or large-sized enterprises (3 %).

3 Estimation strategy

Using the potential outcome approach to causal inference (Rubin 1974, 1978), we estimate a continuous dose-response function that relates each value of the dose, i.e., incentive level, to the post-treatment firm employment gain score. Formally, consider

a set of N enterprises, and denote each of them by subscript i: $i = 1, \ldots, N$. For each enterprise i, we observe a vector of pre-treatment variables, X_i, the received incentive amount, T_i, and the value of the outcome variable associated with this treatment level, $Y_i = Y_i(T_i)$. In our application, the outcome variable, Y_i, is the employment gain score. We use employment gain score instead of employment level as outcome variable in order to remove time-invariant "fixed effects" that might influence the employment level during intervention, and leading to a biased estimate of the treatment effect. Specifically, although the target causal estimand is the same whether using either the employment gain score or the employment level, the employment gain score allows us to remove systematic differences in the time-invariant background characteristics (having an additive effect on Y_i) among firms, receiving different amounts of financial aid.

Alternative outcome variables could be considered. For instance, the relative employment gain score could be used. This variable has the advantage of taking into account the potential relationship between employment gain score and firm size, but the drawback of giving a higher weight to employment gain scores observed in small-sized firms. In addition, from a social-economic standpoint, focus is on the effectiveness of the policies in generating new employment opportunities, and the social benefits of employment growth do not generally depend on firm size.

In order to formally describe the econometric framework we adopt, some additional notation is required. Let $Y_i(t)$ denote a random variable that maps a particular potential treatment, t, to a potential outcome. We are interested in the average dose-response function, $\mu(t) = E[Y_i(t)]$. Following Hirano and Imbens (2004), we assume that $\{Y_i(t)\}_{t\in\mathcal{T}}$, T_i, and X_i, $i = 1, \ldots, N$ are defined on a common probability space, that T_i is continuously distributed with respect to Lebesgue measure on $\mathcal{T}$, and that $Y_i = Y_i(T_i)$ is a well defined random variable.

Note that we make the stable unit treatment value assumption (SUTVA), formalized with potential outcomes by Rubin (1980, 1990). SUTVA combines the no-interference assumption (Cox 1958) with the assumption that there are "no different versions of treatments." In our study, the no-interference assumption requires that the amount of incentive received by one firm should not affect another firm's outcome.[3]

As in the binary treatment context, propensity score methods in a setting with continuous treatments rely on the key assumption that adjusting for pre-treatment differences solves the problem of drawing causal inference. Formally, we make the weak unconfoundedness assumption, introduced by Hirano and Imbens (2004), which requires that the treatment assignment mechanism is conditional independent of each potential outcome given the pre-treatment variables: $Y_i(t) \perp T_i | X_i$ for all $t \in \mathcal{T}$. This assumption implies that all variables which affect both the outcome (employment gain score) and the likelihood of receiving a given amount of financial aid are observed, and that all the others are perfectly collinear with the observed ones. Although this

[3] In our application, the no interference part of SUTVA might be violated due to general equilibrium effects. Alternative assumptions, avoiding the no-interference assumption, could be considered. For instance, interactions between firms could be modeled through assumptions on market structure. The development of approaches where the no-interference assumption is at least partially relaxed is a valuable topic for future research.

assumption is not testable, it is a very strong assumption, and one that does not need to be generally applicable. The plausibility of such an assumption should be discussed taking into account the richness of the information provided by the pre-treatment covariates in relation to a detailed understanding of the institutional set-up where selection into the treatment takes place. In our application we assume that unconfoundedness holds conditional on all the pre-treatment variables listed in Tables 2 and 3, arguing that these variables are good proxies of factors that might affect the intensity of the financial aid (Rubin (2008)). Although this assumption seems quite plausible, we cannot completely rule out the possibility that there exist systematic differences in unobservable characteristics, such as firm's propensity to invest in employment and motivation and ability of the firm to improve its own performances, biasing our estimated effects. If these unobservable characteristics result in firms with higher growth skills systematically receiving lower (higher) amounts of financial aids, our estimated effects would underestimate (overestimate) the true effects. For instance, our estimated effects would be biased downward if firms with higher growth skills propose less expensive projects and receive lower amounts of financial aid, even after controlling for covariates. Conversely, our estimated effects would be biased upward if, for example, even after controlling for covariates, higher incentive levels are given to firms with the best project, leading them to experience a faster growth, and/or if there is a positive correlation between the amount of financial aid asked and the amount of financial aid actually received. Despite of these potential confounders, we view the unconfoundedness assumption as a useful starting point. However, in order to assess the robustness of our results with respect to the underlying identifying assumptions, we also present a simulation-based sensitivity analysis aimed at assessing the sensitivity of estimates to specific deviations from the unconfoundedness assumption. Alternative approaches, such as instrumental variables, could be also considered, although they generally require alternative untestable assumptions, such as exclusion restriction assumptions, which might be even more difficult to justify.

Given unconfoundedness, we can apply methods based on the GPS with continuous treatments introduced by Hirano and Imbens (2004). The GPS is defined as the conditional density of the actual treatment given the observed covariates. Formally, let $r(t, x) = f_{T|X}(t|x)$ be the conditional density of the treatment given the covariates. Then, the GPS is $R_i = r(T_i, X_i)$. The GPS is a balancing score (e.g., Rosenbaum and Rubin 1983a), that is, within strata with the same value of $r(t, x)$, the probability that $T = t$ does not depend on the value of X. Combined with the weak unconfoundedness assumption, this balancing property implies that for every $t \in \mathcal{T}$

$$f_T\left(t|r(t, X_i), Y_i(t)\right) = f_T\left(t|r(t, X_i)\right).$$

As a result, the GPS can be used to eliminate any bias associated with differences in the covariates. Formally, if the assignment to the treatment is weakly unconfounded given pre-treatment variables X_i, then

$$\beta(t, r) = E\left[Y_i(t)|r(t, X_i) = r\right] = E\left[Y_i|T_i = t, R_i = r\right]$$

and

$$\mu(t) = E\left[\beta(t, r(t, X_i))\right].$$

This result suggests that the dose-response function at t can be estimated using a partial mean approach (Newey 1994), that is, averaging the regression function $\mu(t, r(t, X_i))$ over the covariates X_i, while holding fixed the treatment level t. Hence, the dose-response function can be estimated using the GPS by the following two steps. The first involves the estimation of the conditional expectation of Y on T and the estimated GPS, $\widehat{R}$, $E(Y_i|T_i, \hat{R}_i)$, while the second involves averaging this conditional expectation over $\widehat{r}(t, X_i)$ to get the value of the dose-response function at t, where $\widehat{r}(t, X_i)$ are the predicted scores at treatment level t. Note that, we do not average over the GPS, $R_i = r(T_i, X_i)$, but instead we average over the score evaluated at the treatment level of interest, $r(t, X_i)$.

In our application, GPS methods are implemented using the STATA package written by Bia and Mattei (2008). A key assumption in the STATA implemented version of the GPS methods is the normality of the treatment variable (or of its transformation) conditional on the pre-treatment covariates. In our application we assume that the Box-Cox transformation of the treatment (amount of the financial aid) has a Normal distribution, given the covariates. Formally, let $BoxCox(T_i)$ denote the Box-Cox transformation of the treatment variable:[4]

$$BoxCox(T_i) = \begin{cases} \frac{T_i^{\lambda}-1}{\lambda}, & \text{if } \lambda > 0; \\ \log(T_i), & \text{if } \lambda = 0. \end{cases}$$

Then, we assume that

$$BoxCox(T_i)|X_i \sim N\left(b_0 + b_1 X_i, \sigma^2\right).$$

Next, we estimate the dose-response function by assuming a (flexible) parametric form for the regression function of Y on T and R.

4 Results

4.1 Covariate balance after adjustment for the GPS

To be effective, GPS methods should balance characteristics across treatment groups. The extent to which this has been achieved can be explored by comparing balance in the pre-treatment covariates before and after adjusting for the estimated GPS. Specifically, we first divide the observed range of the received incentives into sub-intervals. We apply a different classification strategy of the incentive range

[4] The parameter λ, which the Box-Cox transformation depends on, is estimated from the data. In our study, the estimate of λ is 0.123 (SE = 0.011) for the small-sized firms, and 0.177 (SE = 0.029) for the medium- or large-sized firms.

for small-sized and medium- or large-sized firms. We use seven incentive intervals for small-sized firms: [500; 4,000), [4,000; 5,500), [5,500; 7,000), [7,000; 20,000), [20,000; 50,000), [50,000; 100,000) and [100,000; 200, 000); and five incentive intervals for medium or large-sized firms: [500, 15,000), [15,000; 35,000), [35,000; 75,000), [75,000; 200,000), [200,000; 700,000). Then, for each covariate, we investigate the balance by testing whether the mean in each pre-fixed treatment interval is different from the mean in the other ones.

Tables 4 and 6 provide the unadjusted and GPS—adjusted mean differences and the Bayes—Factor Test statistics for equality of means.[5] Adjusting for the GPS seems to improve the balance, especially when the unadjusted differences are high. For the small-sized firms, after the adjustment for the GPS, 9 Bayes—Factors out of 98 are lower than 1 (compared to 28 prior to adjustment) and none is lower than $10^{-0.5}$ (compared to 21 prior to adjustment). For the medium or large-sized enterprises, after the adjustment for the GPS, only 3 Bayes - Factors out of 60 are lower than 1 (compared to 6 prior to adjustment) and none is lower than $10^{-0.5}$ (compared to 2 prior to adjustment). As an example, consider the province of Turin. The unadjusted mean difference between the first treatment group ([500; 4,000] for small-sized firms and [500; 15,000) for medium- or large-sized ones) and the other ones is $0.169(\text{SE} = 0.025)$ for the small-sized firms, and $0.227(\text{SE} = 0.060)$ for the medium- or large-sized firms. After adjusting for the GPS, these differences reduce to $0.035(\text{SE} = 0.028)$, and $0.001(\text{SE} = 0.041)$, respectively.

4.2 Estimation of the dose-response and treatment functions

Given these results, it is possible to use the GPS to remove bias associated with differences in covariates applying the following two steps. First we estimate the conditional expectation of the outcome, employment gain score, as a function of two scalar variables, the treatment level, T_i, and the estimated GPS, $\hat{R}_i$, using a quadratic approximation:

$$E(Y_i|(T_i, \hat{R}_i) = \alpha_0 + \alpha_1 T_i + \alpha_2 T_i^2 + \alpha_3 \hat{R}_i + \alpha_4 \hat{R_i^2} + \alpha_5 T_i \hat{R}_i.$$

Second we estimate the dose-response function at each specific level of the treatment, by averaging the conditional expectation function over the GPS at that particular level of the treatment, $\mu(t) = E[\beta(t, r(t, X_i))]$.

Figure 1a shows the average dose-response function of the employment gain score for small-sized firms and the 95 % confidence bands. As we can see in Fig. 1a, the employment gain score is an increasing function of the amount of the financial aid received. The average employment gain score appears to be positive for each value of the grant greater than 10,000 Euro, and the confidence bands are quite narrow, except for large incentive levels. For instance, a grant of 10,000 Euro leads to an average

[5] Bayes factors are interpreted following Jeffreys (1961). Let B_{01} denote the Bayes Factor for nested models $M_0 \subset M_1$, then $B_{01} > 1$: evidence supports M_0; $1 > B_{01} > 10^{-0.5}$: very slight against M_0; $10^{-0.5} > B_{01} > 10^{-1}$: moderate against M_0; $10^{-1} > B_{01} > 10^{-2}$: strong to very strong against M_0; $10^{-2} > B_{01}$: decisive against M_0.

Table 4 Small-sized firms. Balance given the Generalized Propensity Score. Mean Difference (MD) (Standard error, SE, in parenthesis) and Bayes Factor statistics for equality of means

Treatment interval	[500; 4000)				[4000; 5500)				[5500; 7000)				[7000; 20000)			
	Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj	
Variables	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF
Province																
Turin	0.169 (0.025)	0.00	0.035 (0.028)	6.05	−0.04 (0.036)	11.26	−0.098 (0.037)	0.28	0.004 (0.034)	21.89	−0.047 (0.034)	3.85	0.01 (0.017	36.4	−0.013 (0.017)	14.72
Vercelli	0.00 (0.009)	30.26	−0.002 (0.011)	13.53	0.014 (0.014)	12.46	0.01 (0.015)	7.47	0.007 (0.013)	18.91	0.014 (0.014)	5.66	−0.011 (0.007)	11.24	−0.011 (0.007)	5.56
Novara	−0.065 (0.014)	0.00	−0.032 (0.015)	1.4	0.019 (0.020)	13.37	0.034 (0.020)	2.25	0.035 (0.019)	3.96	0.048 (0.019)	0.39	0.011 (0.010)	23.3	0.015 (0.010)	6.29
Cuneo	−0.022 (0.015)	10.55	0.004 (0.017)	13.36	−0.024 (0.022)	11.96	−0.009 (0.023)	8.77	−0.01 (0.021)	19.55	−0.002 (0.022)	9.83	−0.007 (0.011)	35.83	−0.001 (0.011)	19.7
Asti	−0.025 (0.011)	1.84	0.002 (0.011)	13.62	0.021 (0.015)	8.16	0.026 (0.015)	2.24	−0.008 (0.015)	18.92	0.009 (0.014)	8.18	−0.006 (0.007)	32.6	0 (0.007)	19.71
Alessandria	−0.066 (0.014)	0.00	−0.028 (0.015)	2.54	−0.019 (0.021)	13.99	−0.003 (0.021)	9.37	0.002 (0.020)	21.92	0.001 (0.020)	9.88	0.016 (0.010)	12.38	0.023 (0.010)	1.44
Biella	0.016 (0.011)	11.40	0.009 (0.014)	11.12	0.025 (0.016)	6.13	0.025 (0.018)	3.56	0.01 (0.015)	18.01	0.002 (0.017)	9.8	−0.001 (0.008)	43.61	−0.006 (0.008)	15.3
Verbano-Cusio-Ossola	−0.007 (0.010)	24.79	0.012 (0.010)	7.3	0.003 (0.015)	20.6	0.014 (0.015)	5.96	−0.039 (0.014)	0.54	−0.025 (0.014)	1.89	−0.013 (0.007)	7.92	−0.007 (0.007)	11.44
Manufacturing industry	−0.01 (0.007)	10.28	−0.003 (0.009)	12.9	−0.001 (0.010)	20.95	0.001 (0.011)	9.37	0.007 (0.010)	17.11	0.011 (0.010)	5.39	0.001 (0.005)	42.15	0.003 (0.005)	15.8

Table 4 continued

Treatment interval	[500; 4000)				[4000; 5500)				[5500; 7000)				[7000; 20000)			
	Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj	
Variables	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF
No handicraft firm	0.207	0.00	−0.007	12.93	0.152	0.00	0.043	2.31	0.143	0.00	0.03	4.73	0.076	0.00	−0.014	10.92
	(0.024)		(0.020)		(0.035)		(0.026)		(0.033)		(0.024)		(0.017)		(0.013)	
Number of employees	5.457	0.00	1.173	0.97	4.506	0.00	2.158	0.04	3.79	0.00	1.512	0.35	3.131	0.00	0.828	0.09
	(0.557)		(0.509)		(0.811)		(0.656)		(0.774)		(0.585)		(0.388)		(0.252)	
Micro enterprise (<10 employees)	−0.22	0.00	−0.002	13.69	−0.103	0.29	0.023	6.04	−0.121	0.03	−0.012	8.78	−0.115	0.00	−0.021	5.27
	(0.024)		(0.019)		(0.035)		(0.024)		(0.034)		(0.024)		(0.017)		(0.013)	
Uni−located firm	−0.041	2.10	0.009	12.66	−0.019	15.83	0.005	9.29	−0.052	2.37	−0.022	7.09	−0.02	11.75	−0.004	18.97
	(0.018)		(0.022)		(0.026)		(0.029)		(0.025)		(0.027)		(0.012)		(0.013)	
Birth year: after 2000	−0.037	1.85	−0.019	7.44	−0.012	18.34	0.007	9.01	0.006	21.29	0.012	8.47	−0.008	32.54	0.005	17.77
	(0.016)		(0.017)		(0.023)		(0.023)		(0.022)		(0.022)		(0.011)		(0.011)	

Table 4 continued

Treatment interval	[20000; 50000)				[50000; 100000)				[100000; 200000)			
	Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj	
Variables	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF
Province												
Turin	−0.034 (0.018)	7.2	−0.014 (0.018)	13.95	−0.058 (0.022)	0.90	−0.015 (0.023)	12.73	−0.042 (0.028)	8.43	0.027 (0.032)	8.56
Vercelli	0.006 (0.007)	26.6	0.008 (0.007)	8.94	0.001 (0.008)	34.37	0.004 (0.009)	14.39	−0.003 (0.011)	26.03	−0.006 (0.013)	10.93
Novara	−0.018 (0.010)	7.96	−0.028 (0.010)	0.49	0.019 (0.012)	10.38	0.004 (0.014)	15.20	0.032 (0.016)	3.42	0.024 (0.019)	5.66
Cuneo	0.024 (0.011)	4.51	0.02 (0.011)	3.94	0.017 (0.013)	15.13	0.016 (0.015)	9.25	−0.018 (0.017)	15.46	−0.036 (0.021)	2.85
Asti	0.003 (0.008)	38.51	0.001 (0.008)	18.56	0.013 (0.009)	13.01	0 (0.011)	15.74	0.01 (0.012)	18.64	0.007 (0.015)	10.88
Alessandria	0.028 (0.010)	1.13	0.022 (0.011)	2.38	−0.008 (0.013)	28.8	−0.027 (0.014)	2.77	−0.003 (0.016)	26.32	−0.036 (0.020)	2.57
Biella	−0.015 (0.008)	7.19	−0.011 (0.008)	7.48	−0.011 (0.010)	18.74	−0.004 (0.010)	14.63	0.014 (0.013)	13.94	0.02 (0.014)	4.51
Verbano-Cusio-Ossola	0.006 (0.008)	31.51	0.001 (0.008)	18.75	0.027 (0.009)	0.49	0.022 (0.011)	1.8	0.01 (0.012)	18.34	0.00 (0.015)	12.2
Manufacturing industry	−0.001 (0.005)	41.01	−0.003 (0.005)	15.59	0.003 (0.006)	31.13	−0.002 (0.006)	15.19	0.003 (0.008)	24.80	0.002 (0.009)	11.99

Table 4 continued

Treatment interval	[20000; 50000)				[50000; 100000)				[100000; 200000)			
	Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj	
Variables	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF
No handicraft firm	−0.048	0.96	0.027	2.67	−0.182	0.00	−0.047	0.63	−0.232	0.00	−0.086	0.08
	(0.018)		(0.014)		(0.021)		(0.019)		(0.027)		(0.027)	
Number of employees	−2.023	0.00	−0.613	3.48	−4.805	0.00	−0.342	6.83	−7.636	0.00	−0.765	1.46
	(0.407)		(0.334)		(0.488)		(0.264)		(0.624)		(0.371)	
Micro enterprise (<10 employees)	0.105	0.00	0.029	1.80	0.155	0.00	0.001	15.69	0.216	0.00	0.014	10.76
	(0.018)		(0.014)		(0.021)		(0.018)		(0.027)		(0.027)	
Uni-located firm	0.014	23.05	0.001	18.82	0.036	2.52	0	15.74	0.059	0.39	0.029	4.72
	(0.013)		(0.013)		(0.016)		(0.016)		(0.020)		(0.021)	
Birth year: after 2000	0.008	33.56	0.003	18.35	0.029	4.09	0.018	8.19	0.007	24.69	−0.017	9.11
	(0.011)		(0.012)		(0.014)		(0.016)		(0.018)		(0.022)	

Table 5 Medium- or large-sized Firms. Balance given the Generalized Propensity Score. Mean Difference (MD) (Standard error, SE, in parenthesis) and Bayes Factor statistics for equality of means

Treatment Interval	[500; 15000)				[15000; 35000)				[35000; 75000)				[75000; 200000)				[200000; 700000)			
	Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj	
Variables	MD f(SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF (SE)	MD	BF (SE)	MD	BF
Province																				
Turin	0.227	0.01	0.001	5.68	0.084	3.79	0.056	1.36	0.036	11.90	−0.031	3.74	−0.018	16.41	−0.007	7.62	−0.224	0.00	0.038	1.73
	(0.060)		(0.041)		(0.051)		(0.032)		(0.046)		(0.027)		(0.042)		(0.022)		(0.046)		(0.023)	
Vercelli	−0.009	11.72	−0.029	3.14	−0.011	12.60	−0.029	2.75	0.013	12.74	0.014	5.49	−0.001	17.95	0.002	7.92	0.002	15.87	0.002	7.35
	(0.025)		(0.027)		(0.021)		(0.022)		(0.019)		(0.019)		(0.017)		(0.016)		(0.019)		(0.023)	
Novara	−0.110	0.15	−0.002	5.67	−0.041	6.44	0.001	6.52	−0.050	3.51	−0.042	2.23	0.027	10.22	0.015	5.96	0.118	0.00	0.086	0.49
	(0.037)		(0.022)		(0.032)		(0.017)		(0.029)		(0.027)		(0.026)		(0.020)		(0.029)		(0.037)	
Cuneo	−0.025	9.94	0.028	3.78	−0.011	13.77	−0.009	6.20	−0.001	16.24	0.030	3.23	−0.030	8.95	−0.032	3.23	0.063	1.44	−0.017	6.60
	(0.037)		(0.032)		(0.032)		(0.027)		(0.028)		(0.024)		(0.026)		(0.024)		(0.029)		(0.037)	
Asti	−0.022	7.34	−0.031	2.17	0.017	9.34	0.016	4.46	−0.002	16.15	−0.001	7.29	0.011	13.84	0.012	5.76	−0.013	11.94	−0.022	4.04
	(0.021)		(0.023)		(0.018)		(0.019)		(0.016)		(0.016)		(0.015)		(0.015)		(0.017)		(0.020)	
Alessandria	−0.027	8.14	−0.002	5.67	−0.006	14.12	−0.008	6.18	0.015	12.92	0.034	1.94	0.002	17.85	0.001	7.98	0.003	15.82	−0.056	1.26
	(0.030)		(0.027)		(0.025)		(0.024)		(0.023)		(0.021)		(0.021)		(0.020)		(0.023)		(0.030)	
Biella	0.010	11.98	0.035	3.32	−0.019	11.94	−0.027	4.38	−0.019	12.96	−0.007	6.99	0.001	17.95	0.004	7.88	0.028	9.81	−0.042	3.62
	(0.036)		(0.034)		(0.031)		(0.030)		(0.028)		(0.025)		(0.025)		(0.022)		(0.028)		(0.035)	
Verbano-Cusio-Ossola	−0.044	0.73	−0.001	5.67	−0.013	10.58	−0.002	6.49	0.006	14.87	0.003	7.15	0.007	15.36	0.005	7.51	0.022	4.92	0.011	6.39
	(0.019)		(0.011)		(0.016)		(0.013)		(0.014)		(0.015)		(0.013)		(0.013)		(0.015)		(0.020	
Number of employees	21.263	3.45	−0.318	5.68	12.857	7.67	12.674	3.03	12.362	7.78	12.526	3.17	−22.651	0.86	−23.983	0.23	−7.804	12.03	−3.796	6.73
	(13.265)		(16.127)		(11.362)		(10.277)		(10.193)		(9.743)		(9.185)		(8.939)		(10.380)		(8.882)	

Table 5 continued

Treatment Interval	[500; 15000)				[15000; 35000)				[35000; 75000)				[75000; 200000)				[200000; 700000)			
	Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj		Un-Adj		GPS-Adj	
Variables	MD f(SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF	MD (SE)	BF (SE)	MD	BF (SE)	MD	BF
Medium-sized enterprise	−0.029	7.74	−0.001	5.68	−0.024	9.40	−0.023	4.19	0.003	16.06	−0.002	7.25	0.048	1.07	0.049	0.41	−0.028	7.69	−0.038	2.26
(50–250 employees)	(0.029)		(0.033)		(0.025)		(0.025)		(0.023)		(0.022)		(0.020)		(0.020)		(0.023)		(0.025)	
Uni-located firm	0.020	11.82	−0.006	5.66	−0.046	9.74	−0.056	3.49	−0.032	12.63	−0.029	5.86	0.039	11.46	0.039	4.97	0.010	15.59	0.006	7.34
	(0.059)		(0.066)		(0.051)		(0.051)		(0.046)		(0.045)		(0.041)		(0.040)		(0.046)		(0.051)	
Birth year: after 2000	−0.025	5.78	−0.005	5.39	0.003	14.36	0.006	6.00	−0.014	10.82	−0.008	6.29	0.000	17.96	−0.003	7.83	0.028	3.35	0.015	5.68
	(0.020)		(0.016)		(0.017)		(0.015)		(0.016)		(0.015)		(0.014)		(0.014)		(0.016)		(0.021)	

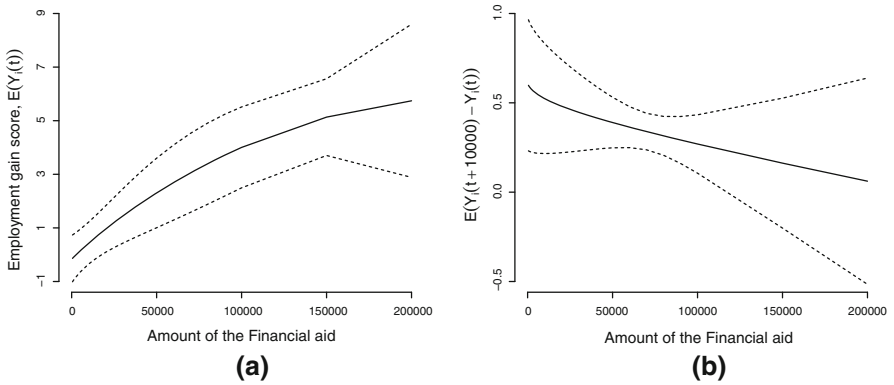


Fig. 1 Small-sized firms: **a** Average dose-response function and 95 % confidence bands for the employment gain score; **b** Average effect of an incentive gap of 10,000 Euro on employment gain score and 95 % confidence bands. The confidence bands are based on 100 bootstrap replications, taking into account estimation of the GPS

employment gain score of 0.43 (SE = 0.40), and a grant of 20,000 Euro leads to an average employment gain score of 0.96 (SE = 0.44). This implies that the average causal effect of a grant of $t = 20{,}000$ Euro relative to a grant of $s = 10{,}000$ Euro is $0.96-0.43 = 0.53$ (SE = 0.16) extra workers, so that, increasing the amount of financial aid from 10,000 to 20,000 Euro redoubles the number of new workers, as we can also see in Fig. 1b, which shows the average effect of an incentive gap of 10,000 Euro on employment gain score. Figure 1b stresses that marginal effects are larger for low levels of the financial aid. For instance, a grant of 10,500 Euro relative to a grant of 500 Euro leads to hiring 0.60(SE = 0.19) extra workers, but only 0.27(SE = 0.08) extra workers are recruited due to increasing the amount of financial aid from 100,000 to 110,000 Euro. The estimated marginal effects of grants greater than 100,000 Euro are very small and statistically negligible: the estimated average treatment effect function takes on values around zero, and the corresponding confidence bands are somewhat wide and cross zero. Thus, there appears to be some evidence that firms receiving lower incentive levels are much more sensitive to incentive changes than firms receiving higher incentive levels.

Figure 2a shows the average dose-response function along with 95 % confidence bands for medium- or large-sized firms. Although the 95 % confidence bands are somewhat large, implying a high level of uncertainty on the shape of the average dose-response function, our findings may provide some useful insights for policy makers. Specifically, there is some evidence that the average employment gain score increases from 3.61 (SE = 7.57) to 22.37 (SE = 9.54) as the amount of the financial aid increases from 500 to 300,000 Euro. The average employment gain score slightly decreases as the amount of the financial aid increases from 300,000 to 500,000 Euro, and goes down roughly for incentive levels greater than 500,000 Euro. As far as the effect of an incentive gap of 10,000 Euro on employment is concerned, Fig. 2b suggests that increasing the amount of the financial aid by 10,000 Euro has a small and in general negligible effect on employment. The average treatment function is increasing

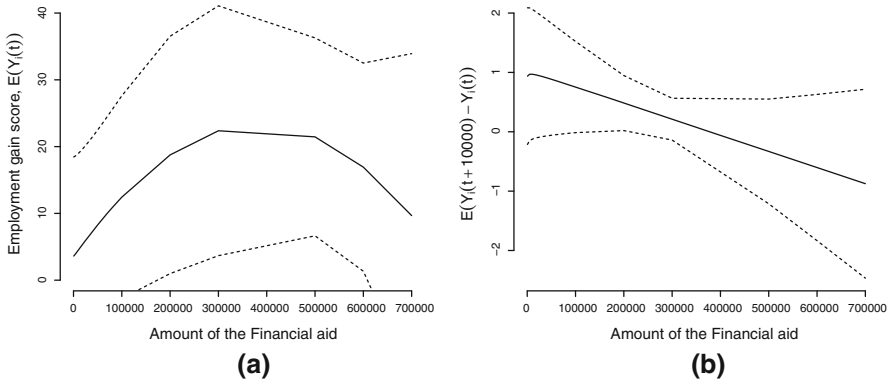


Fig. 2 Medium- or large-sized firms: **a** Average dose-response function and 95 % confidence bands for the employment gain score; **b** Average effect of an incentive gap of 10,000 Euro on employment gain score and 95 % confidence bands. The confidence bands are based on 100 bootstrap replications, taking into account estimation of the GPS

for incentive levels lower than 20,000, and decreasing as the amount of the financial aid increases from 20,000 to 700,000 Euro, reaching also negative values for incentive levels greater than (or equal to) 500,000 Euro. Therefore, there is some evidence that increasing the amount of financial aid may push medium or large-sized firms to hire some extra workers for incentive levels lower than 500,000 Euro, although most of the estimated marginal effects are not significant at the 5 % level.

It could be argued that an incentive gap of 10,000 Euro is irrelevant for medium- or large-sized enterprises, which on average receive incentives greater than 100,000 Euro. Therefore, we also estimate the average treatment effect function considering an incentive gap of 50,000 Euro. As can be seen by comparing Figs. 2b and 3, the behaviour of the effect of increasing the amount of the financial aid by 50,000 Euro is similar to the behaviour of the effect of increasing the amount of the financial aid by 10,000 Euro, although an incentive gap of 50,000 Euro has greater effects. Specifically, the marginal effect of increasing the amount of the financial aid by 50,000 Euro decreases monotonically, reaching also negative values.

These estimated effects are important for policy, providing useful insights into the possibility to find the optimal level of incentives, that is, the incentive level at which the average dose-response function is maximized, and to estimate the maximum value achieved by the average dose-response function. Following Flores (2007), we name the treatment level at which that average dose-response function is maximized 'location of the optimal dose', and the maximum value achieved by that curve 'size of the optimal dose.' In our study, it may be of great interest for policy makers to know the level of contributions that maximizes the average net employment benefits of the financial measures we focus on. For medium- or large-sized firms, the estimated average dose-response function suggests that there exist an inverted U-shaped relationship between incentive levels and employment gain score with a turning point around 300,000 Euro (see Fig. 2a). The estimated employment gain score at this point is 22.37, with a standard error of 9.54. This result provides valuable information to policy makers regarding

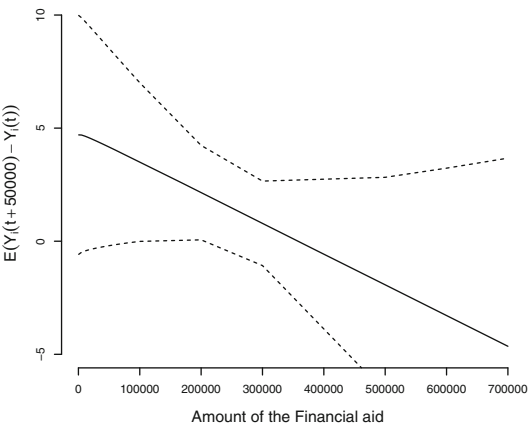


Fig. 3 Medium- or large-sized firms: Average effect of an incentive gap of 50,000 Euro on employment gain score and 95 % confidence bands. The confidence bands are based on 100 bootstrap replications, taking into account estimation of the GPS

the optimal level of incentives to apply for medium- or large-sized firms, suggesting that too low incentive levels may be ineffective, but extremely high incentive levels may be less effective than medium-high incentive levels. For small-sized firms, the estimated dose-response function suggests an increasing relation between employment gain score and level of financial aids, making difficult to find the optimal level of incentives. However, although the average dose-response function is increasing in the amount of financial aid, its curvature seems to weaken for incentive levels greater than 150,000 Euro, therefore we could consider incentive levels around 150,000 Euro as optimal doses (see Fig. 1).[6]

Generally speaking, the key to a successful policy is to get empirical evidence to establish effective strategies supporting firms and local labour markets. From a social perspective, each new employment opportunity is considered as a "tool" that will provide long-term help to stabilize and support local communities. Thus, the marginal benefit of each additional worker may be high from a social standpoint. Moreover, supporting small business is also likely to be a good strategy: "small business can grow to become the large business of tomorrow," offering all the benefits that generally the "recruitment" in a large firm implies (Edmiston 2007). The measures we consider aim at boosting the local socio-economic development, and we find positive and significant effects at least for small-sized firms. Therefore, our results suggest that public policies should be aimed at facilitating the productivity process and speeding up the economic growth by allocating funds to support firms, especially small-sized firms, which seem to strongly benefit from these type of interventions.

[6] This discussion is based on the estimated dose-response functions, and provides only some insights on the possibility to find the location and size of the optimal dose. Flores (2007) proposes a non-parametric method to estimate the optimal dose with continuous treatment: applying this method in our setting may be a valuable topic for future research.

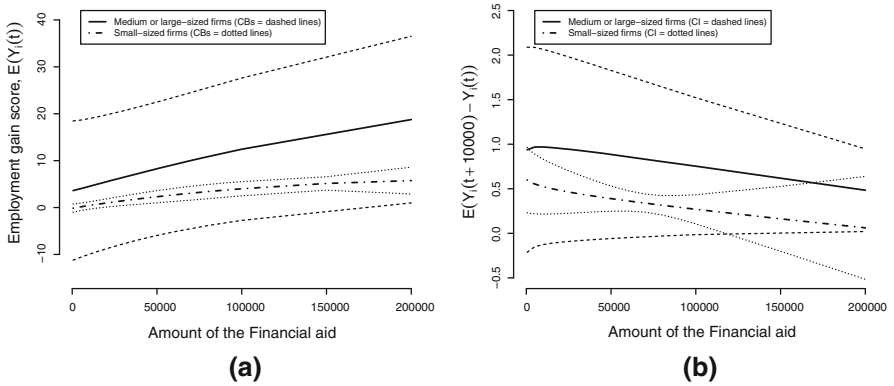


Fig. 4 Small-sized versus medium- or large-sized firms: **a** Average dose-response function and 95 % confidence bands for the employment gain score; **b** Average effect of an incentive gap of 10,000 Euro on employment gain score and 95 % confidence bands. The confidence bands (CBs) are based on 100 bootstrap replications, taking into account estimation of the GPS

4.3 Heterogeneity in the dose-response and treatment effect functions

Although our results suggest that treatment effects are heterogeneous with respect to firm size, by comparing Figs. 1 to 2, we can note that the shapes of the dose-response and treatment effect functions for small-sized firms and medium- or large-sized firms appear to be quite similar for treatment levels lower than 200,000 Euro. In order to better investigate the heterogeneity of these functions across firms' size in this treatment interval, Fig. 4 superimposes the dose-response and treatment-effect functions for small-sized firms on those for medium- or large-sized firms (along with their 95 % confidence bands). Although the two dose-response and treatment-effect functions are indeed quite similar for treatment levels lower than 200,000 Euro, the confidence bands on the dose-response and treatment-effect functions for medium- or large-sized firms are much larger than those for small-sized firms. In addition, statistically significant results are found only for small-sized firms. These differences in the estimated dose-response and treatment functions between small-sized firms and medium- or large-sized firms might be due to various reasons. For instance, it is possible that medium- or large-sized firms are more willing to invest on projects that mainly focus on improving labour efficiency, by leading consequently to a shrinking of the labor force. Conversely, labour-saving innovation might be a minor issue for small-sized firms, which might prefer to spend more in scaling up operations and increase their labour demand. In order to gain some insight on the propensity of different firms to invest in employment, we investigate heterogeneities in the effects of the amount of financial aid with respect to observable characteristics for small-sized firms.[7]

As we can see in Fig. 5, our estimates show important differences in the estimated treatment effects across different groups of firms. The average treatment effects for

[7] We focus only on small-sized firms because the smaller sample size for medium- or large-sized firms makes difficult to investigate heterogeneities in the treatment effects for this group.

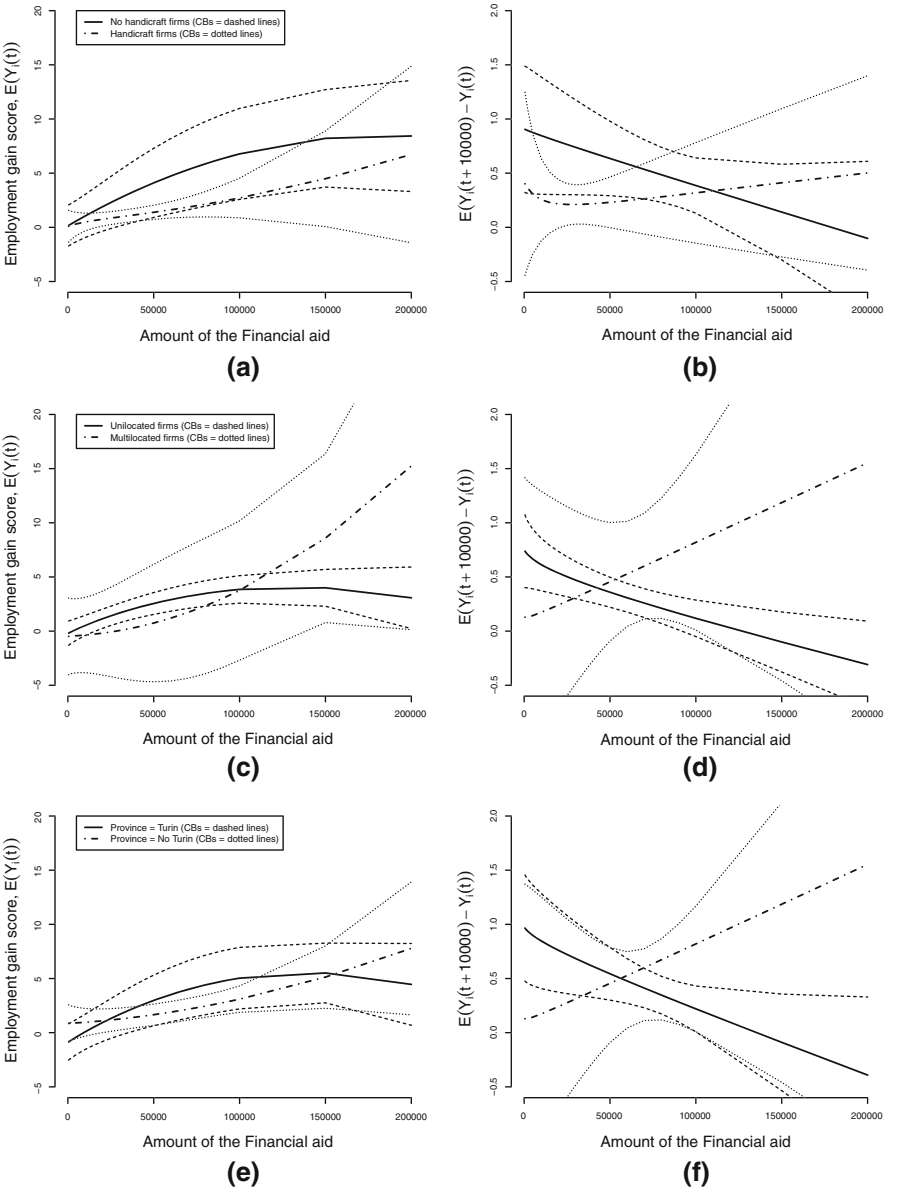


Fig. 5 Small-sized firms: **a**, **c**, **e** Average dose-response functions and 95 % confidence bands for the employment gain score; **b**, **d**, **f** Average effects of an incentive gap of 10,000 Euro on employment gain score and 95 % confidence bands. The confidence bands (CBs) are based on 100 bootstrap replications, taking into account estimation of the GPS

handicraft firms, multilocated firms, and firms with registered office out of Turin are positive and increasing with incentive level, although the confidence bands are somewhat wide, indicating that very few estimates are statistically different from zero. It is worth to note, however, that non-negative effects cannot be ruled out throughout,

because none of the estimates are statistically negative. The average treatment effects for no-handcraft firms, unilocated firms, and firms with registered office in Turin are similar to those of the full sample of small-sized firms (see Fig. 1). Specifically, the average treatment effects for these groups are diminishing with amount of financial aid, also taking on negative values for high incentive levels, although none of the estimates is statistically negative. These results suggest that there might exist important differences in the use of financial aids across firms. For instance, some firms may prefer investments that allow them to bring up labor-saving innovations, whereas some other firms may primarily aim at scaling up operations, by also increasing their labor force.

4.4 Sensitivity to unconfoundedness

All the previous results are based on the plausibility of the identifying assumption of unconfoundedness, which may be a questionable assumption. In order to assess the robustness of our results with respect to violations of this assumption, we conclude this section by performing a sensitivity analysis aimed at indirectly assessing the unconfoundedness assumption. We investigate whether our estimates can be changed substantially when unconfoundedness is assumed to fail in some meaningful ways, which are interpreted as evidence of the presence of unobserved covariates that are correlated both with the potential outcomes and with the treatment variable. Note that, since unconfoundedness is not directly testable, sensitivity analyses showing favorable evidence do not imply that unconfoundedness holds, although unfavorable evidence against this assumption casts considerable doubt on scientific validity of the results obtained under the maintained assumption of unconfoundedness.

Various alternative approaches for sensitivity analyses has been developed (e.g., Rosenbaum and Rubin 1983b; Rosenbaum 1987, 2002; Imbens 2003; Ichino et al. 2008). Here we mainly build on the work by Ichino et al. (2008), aiming at assessing the sensitivity of our results with respect to plausible assumptions about an unobserved binary covariate U, but generalizing their approach to account for the continuous nature of our treatment variable. The development of alternative and principled approaches for sensitivity analyses aimed at checking for the presence of unobservable heterogeneity sources and analysing the robustness of the results of evaluation studies with continuous treatment with respect to the underlying identifying assumptions is a valuable topic for future research.

Commonly to other sensitivity analysis methods, we first relax the unconfoundedness assumption slightly, assuming that the treatment assignment mechanism is unconfounded conditional on both the set of observable variables, X, and the binary unobserved covariate, U, which we assume summarizes some unobservable firms' characteristics (such as propensity to invest in employment, motivation and ability): $Y_i(t) \perp T_i | X_i, U_i$ for all $t \in \mathcal{T}$. Then we derive estimates of the average dose-response and treatment functions under different possible scenarios of deviations from unconfoundedness. Specifically, our sensitivity analysis corresponds to fix the parameter of the conditional distribution of the confounding factor U given T and Y, draw a value of U from this distribution for each firm and re-estimate the average dose-response

function and the treatment function by including the simulated U in the set of matching variables. By changing the assumptions about the distribution of U, we can assess the robustness of our results with respect to different hypotheses regarding the nature of the confounding factor.

Following Ichino et al. (2008), we assume that the conditional distribution of the unobserved binary confounding factor U given T and Y does not depend on the observed covariates: $Pr(U = 1|T, Y, X) = Pr(U = 1|T, Y)$, and we measure "a-posteriori" how different values for the sensitivity parameters chosen to simulate U translate into associations of U with Y and T, conditional on X, using partial correlation coefficients. We also assume that the conditional distribution of U given T and Y is described by the following logit model:

$$Pr(U = 1|T = t, Y = y) = \frac{\exp(\alpha_0 + \alpha_1 t + \alpha_2 y + \alpha_3 ty)}{1 + \exp(\alpha_0 + \alpha_1 t + \alpha_2 y + \alpha_3 ty)}.$$

Given this setting, the sensitivity analysis proceeds by first fixing the α parameters, and drawing a value of U for each firm, according to the amount of financial aid the firm received and its employment gain score value. Then U is treated as any other observed covariate, and included in the set of matching variables used to estimate the generalized propensity score, and the average dose-response function and the treatment function. For each given set of values of the sensitivity parameters, α, we repeat the analysis $m = 10$ times and obtain an estimate of the average dose-response function and the treatment function, which are means of the average dose-response functions and the treatment functions over the distribution of the simulated U.

Table 6 and Fig. 6 show the results of the sensitivity analysis for small-sized firms.[8] Each row of the first four columns of Table 6 contains the four sensitivity parameters, α_j, $j = 0, 1, 2, 3$, which characterize the binary distribution of U under which the analysis has been conducted. The last two columns show the partial correlation coefficients between U and Y (employment gain score) given T and X (column 5), and the partial correlation coefficients between U and T (amount of financial aid) given X (column 6), which measure the degree of association between U and Y, and U and T, respectively, while controlling for the effect of the observed covariates X.

Each row of Table 6 assumes that the distribution of U is in turn comparable to the distribution of observables like province (Turin), manufacturing industry, no

[8] The confidence bands for the average dose-response functions and the treatment functions when the unobserved confounder factor U is included in the set of matching variable are calculated by taking into account both the sampling variability, using bootstrap methods, and the uncertainty due to imputing the missing confounder factor U, using a multiple imputation approach (Rubin 1987). Specifically, let m be the number of imputations of U, and let $\widehat{E}_\ell(Y(t))$ and $\widehat{V}_\ell$ be the value and the estimated variance of the estimator of $E(Y(t))$ in the ℓ-imputed data set, $\ell = 1, 2, \ldots, m$. The m statistics are combined to produce the final estimate of $E(Y(t))$ by averaging $\widehat{E}_\ell(Y(t))$ over the m imputations: $\widehat{E}(Y(t)) = m^{-1} \sum_{\ell=1}^{m} \widehat{E}_\ell(Y(t))$, and the total variability associated with $\widehat{E}(Y(t))$ is estimated as $\frac{1}{m} \sum_{\ell=1}^{m} \widehat{V}_\ell + \frac{m+1}{m(m-1)} \sum_{\ell=1}^{m} (\widehat{E}_\ell(Y(t)) - \widehat{E}(Y(t)))^2$. The treatment functions and their standard error are estimated in a similar way.

Table 6 Sensitivity analysis for small-sized firms

Confounder-like	Sensitivity parameters				Partial correlation coefficients	
	α_0	α_1	α_2	α_3	$\rho_{U,Y\|T,X}$	$\rho_{U,T\|X}$
Province: Turin	0.105	0.127	0.003	−0.025	−0.007	0.050
Manufacturing industry	3.903	−0.112	−0.077	0.019	−0.024	−0.067
No handicraft firm	0.408	0.515	0.080	−0.027	0.036	0.214
Micro enterprise	−0.372	−0.583	0.444	−0.028	0.173	−0.209
Uni-located firm	1.715	−0.148	−0.068	−0.021	−0.056	−0.078
Birth year: after 2000	−2.064	−0.126	0.291	−0.048	0.116	0.003

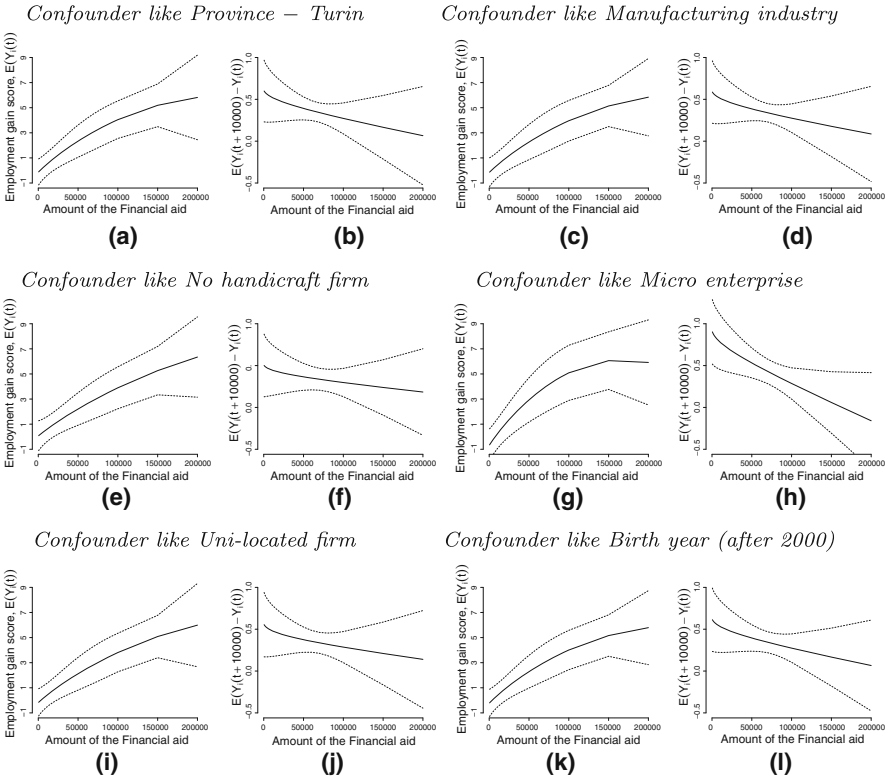


Fig. 6 Small-sized firms—sensitivity analysis: for each set of the sensitivity parameters, a value of the unobserved confounder factor U is imputed to each firm and the average dose-response functions and 95 % confidence bands for the employment gain score (*graphs* **a**, **c**, **e**, **g**, **i**, **k**) and the average effects of an incentive gap of 10,000 Euro on employment gain score and 95 % confidence bands (*graphs* **b**, **d**, **f**, **h**, **j**, **l**) are estimated with U included in the set of matching variable. The process is repeated $m = 10$ times

handicraft firm, micro enterprise, uni-located firm, and birth year. Specifically, for each observed binary covariate, X, we estimate the coefficients of a logit model linking the probability that the covariate takes on value 1 to the treatment variable, T,

the outcome, Y, and their interaction.[9] The estimated values of these model coefficients are in turn used as sensitivity parameters. Choosing the sensitivity parameters α_j, $j = 0, \ldots, 3$ to make the distribution of U similar to the empirical distribution of observable binary covariates allows us to investigate the extent to which matching results are robust to deviations from the unconfoundedness assumption induced by the impossibility of observing factors similar to the ones used to calibrate the distribution of U. Alternative configurations of the sensitivity parameters could be used. However, we believe that the observed associations between observables, treatment and outcome may suggest reasonable and meaningful assumptions on the association between unobserved confounding factors, treatment and potential outcomes.

For each of the simulation scenarios, Fig. 6 shows the estimated average dose-response and treatment functions. The results of the sensitivity analysis convey an impression of robustness of our estimates with respect to reasonable failures of the unconfoundedness assumption: each graph in Fig. 6 appears to be very similar to the corresponding graph in Fig. 1. Only in the case of a confounder factor behaving like micro enterprise, which is also characterized by the strongest association both with Y ($\rho_{U,Y|T,X} = 0.173$) and with T ($\rho_{U,T|X} = 0.209$), the average dose-response function and the treatment function seem to be slightly different from those in Fig. 1. Specifically, for incentive levels larger than 100,000 Euro, the slopes of the curves in graphs (g) and (h) of Fig. 6 appear to be stronger than those of the curves in graphs (a) and (b) of Fig. 1. However, only few small-sized firms receive incentive levels larger than 100,000 Euro, and the causal effects corresponding to contributions larger than 100,000 Euro are still small and statistically negligible.

Table 7 and Fig. 7 present the results of the sensitivity analysis for medium- or large-sized firms. As can be seen in Table 7 and Fig. 7, also for medium- or large-sized firms inferential conclusions are relatively insensitive over the range of configurations of the sensitivity parameters that mimic the distribution of observables (that is, province, medium-sized enterprise, uni-located firm, and birth year). The presence of the confounder factor, U, seems to mainly affect the precision of the estimates for the average dose-response function, leading to somewhat wide confidence intervals. However, large confidence intervals for the average dose-response function are estimated also when no confounder is included in the set of matching variables (see Fig. 2), so the sensitivity analysis does not clash with the evidence provided by the data.

In summary, our sensitivity analysis shows that the presence of unobserved confounders behaving like observables does not change the results substantially. Therefore, although the sensitivity analysis does not prove that unconfoundedness holds, which is an intrinsically non-testable assumption, it provides valuable evidence on the reliability of our estimates.

[9] The treatment and outcome variables are standardized to have zero mean and unit variance before fitting the logit models.

Table 7 Sensitivity analysis for medium- or large-sized firms

Confounder-like	Sensitivity parameters				Partial correlation coefficients	
	α_0	α_1	α_2	α_3	$\rho_{U,Y\|T,X}$	$\rho_{U,T\|X}$
Province: Turin	0.114	0.560	−0.222	−0.026	−0.121	0.217
Medium-sized enterprise	2.727	0.025	−0.315	0.028	−0.049	−0.043
Uni-located firm	0.324	0.027	−0.216	0.052	−0.079	0.017
Birth year: after 2000	−3.660	−0.666	−0.115	0.174	−0.029	−0.055

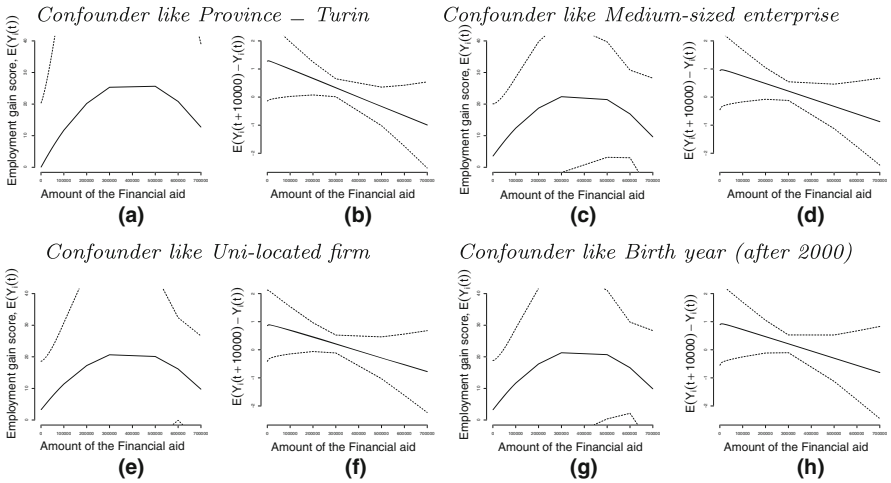


Fig. 7 Medium- or large-sized firms—sensitivity analysis: for each set of the sensitivity parameters, a value of the unobserved confounder factor U is imputed to each firm and the average dose-response functions and 95 % confidence bands for the employment gain score (*graphs* **a**, **c**, **e**, **g**) and the average effects of an incentive gap of 10,000 Euro on employment gain score and 95 % confidence bands (*graphs* **b**, **d**, **f**, **h**) are estimated with U included in the set of matching variable. The process is repeated $m = 10$ times

5 Conclusions

The role of policy makers in management of economic interventions for industries has amplified over the past few years. Empirical evidence is needed in order to establish a correct future evaluation and efficient programs to support companies.

In this paper we address these issues focusing on the causal effect of receiving different amounts of financial aid to firms. We argue that estimating marginal effects of the intensity of economic supports provide more information regarding the effectiveness of the public interventions and might help policy makers in the allocation of public resources.

We estimate the average dose-response functions and the effect of the amount of economic aids on employment disjointedly for small-sized and medium- or large-sized firms, applying propensity score methods, as extended and generalized in a setting with a continuous treatment by Hirano and Imbens (2004).

Our results show a positive and highly significant effect of the amount of the incentive for small-sized firms receiving a financial aid lower than 100,000 Euro. However, lower incentive level appears to be the most effective to improve the employment level: the treatment effect in terms of employment gain score goes down sharply with the level of financial aid, suggesting that firms receiving lower incentive levels are much more sensitive to incentive changes than firms receiving higher incentive levels. For medium- or large-sized enterprises we found some evidence that the intensity of the financial aid has a positive, although decreasing and not highly significant, effect on employment for incentive levels lower than 300,000 Euro. These results also provides some insights into the possibility to find the optimal level of incentives. For small-sized firms, the estimated average dose-response function is always increasing, but less in correspondence of treatment levels higher than 150,000 Euro, suggesting a hypothetical optimal level of incentives around this point. For medium- or large-sized firms, incentive levels around 300,000 Euro seem to have the highest net benefits on employment: in fact, the estimated average dose-response function has an inverted U-shape, with turning point around 300,000 Euro. Although these results rest on the plausibility of the identifying assumption of unconfoundedness, sensitivity analyses convey an impression of robustness of our estimates with respect to reasonable failures of this assumption.

There are two main directions for future research. The first is to extend this analysis by distinguishing between grants and below-market-rate loans, using a mixed continuous treatment variables, in order to jointly estimate the combined effects of different kinds of financial aid. Note that, whereas Bondonio (2008) analyzes the impact of grants and below-market-rate loans separately, we aim at using a multivariate treatment variable, consisting of a discrete part with multiple factors (e.g., grants and below-market-rate loans), and of a continuous part specifying the incentive level. Secondly, we aim at relaxing the normality assumption of the conditional distribution of the treatment given the covariates and the parametric form of the dose-response function, exploring nonparametric regression estimators.

Acknowledgements We are particularly grateful to Fabrizia Mealli and Guido Imbens for their insightful suggestions and advices and Daniele Bondonio for providing the data. We also thank the anonymous Referees for their detailed comments that greatly improved the paper. This research is part of the "Estimation of direct and indirect causal effects using semi-parametric and non-parametric methods" project supported by the Luxembourg "Fonds National de la Recherche" cofunded under the Marie Curie Actions of the European Commission (FP7-COFUND).

Appendix

See Table A.

Table A National, adopted by Region and Regional Policy Measures and DOCUP Measures

Law	Type of measure	Type of contribution
National policy measures		
Law 228/97 art. 4	Economic support for flood risk areas	Interest account contribution
Law 226/99 art. 3	Renegotiation loans L. 35/95 for firms damaged by flood	Interest account contribution
Law 662/96 art. 2, c. 3	Territorial Agreements	Capital account contribution
Law 95/95 (ex 44/86)	"Junior" entrepreneurs	Capital/Working account contribution; Soft loan
Law 236/93	Continuous training	Capital account contribution
Law 488/92 Industry art. 1, c. 2	Productive activities in depressed areas	Capital account contribution
Law 297/99 D.M.593/00	Research funds (FAR)	Grant/ Tax credit
215/92	Female entrepreneurship	Capital account contribution
Policy measures adopted by regions		
Law 598/94 art. 11	Investments in innovation and environmental safeguard	Interest account contribution; Soft Loan
Law 1329/65	"Sabatini" Law	Interest account contribution
Law 341/95 art. 1	Economic support to depressed areas	Tax credit
Law 140/97	R&D	Tax credit
Law 266/97 art. 14	Economic support to companies in depressed areas	Capital account contribution
Law 266/97 art. 8	Economic support to regions	Tax credit
Regional policy measures		
R.L. 67/94	Economic support to employment in cooperatives	Capital account contribution; Soft loan
R.L. 24/97 art. 6	Economic support to industrial development	Capital account contribution
R.L. 22/97	Promotion and economic support to new entrepreneurship	Capital account contribution; Soft loan
R.L. 56/86	Promotion of technological innovation; Interventions for quality-systems	Soft-loan
DOCUP 2000-06 Ob. 2 areas		
Measure 1.2a	International promotion of enterprises	Capital account contribution
Measure 2.1b	Economic support to investments for enterprises	Soft-loan
Measure 2.1d	Economic support to investments related to BEI loans	Capital account contribution
Measure 2.2b	Financial contributions for shared loans	Financial contributions
Measure 2.2c	Strategic advice for enterprise qualification	Capital account contribution
Measure 2.4c	Economic support for e-business	Capital account contribution
Measure 4.1b	Financial contributions to economic activities	Soft-loan

References

Abadie A (2005) Semiparametric differences-in-differences estimators. R Econ Stud 72:1–19

Adorno V, Bernini C, Pellegrini G (2007) The impact of capital subsidies: new estimations under continuous treatment. Giornale degli Economisti e Annali di Economia 66(1):67–92

Adorno V, Bernini C, Pellegrini G (2008) Comparing continuous treatment matching methods in policy evaluation. In: Ingrassia S, Rocci R (eds) Seventh meeting of the CLAssification and data analysis group of the Italian Statistical Society. Short paper, pp 461–464

Battistin E, Gavosto A, Rettore E (2001) Why do subsidised firms survive longer? An evaluation of a program promoting youth enterpreneurship in Italy. In: Lechner M, Pfeiffer F (eds) Econometric evaluation of active labour market policies. Physica/Springer, Heidelberg

Bia M, Mattei A (2008) A stata package for the estimation of the dose Response function through adjustment for the generalized propensity score. Stata J 8(3):354–373
Bondonio D (2002) Evaluating the employment impact of business incentive programs in EU disadvantaged areas. A case from Northern Italy. Working paper, Department of Public Policy and Public Choice POLIS
Bondonio D (2008) La valutazione integrata delle diverse tipologie di aiuto. In: De Blasio G, Lotti e F (eds) La valutazione degli aiuti alle imprese. Il Mulino, Bologna, pp 185–225
Bondonio D, Greenbaum R (2006) Do business investment incentives promote employment in declining areas? Evidence from EU objective 2 regions. Eur Urban Reg Stud 13(3):225–244
Bronzini E, De Blasio G, Pellegrini G, Scognamiglio A (2008) The effect of investment tax credit: evidence from an atypical programme in Italy. Economic working paper, 661, Bank of Italy, Economic Research Department
Cox DR (1958) Planning of experiments. Wiley, New York
Edmiston K (2007) The role of small and large businesses in economic development. Fed Reserv Bank Kans City Econ Rev 92(2):73–97
Flores CA (2007) Estimation of dose-response functions and optimal doses with a continuous treatment, Working Paper, University of Miami
Flores CA, Flores-Lagunes A, Gonzalez A, Neumann TC (2011) Estimating the effects of length of exposure to a training program: the case of job corps. Rev Econ Stat 94(1):153–171
Heckman JJ, Ichimura H, Todd P (1998) Matching as an econometric evaluation estimator. Rev Econ Stud 65:261–294
Hirano K, Imbens G (2004) The propensity score with continuous treatment. In: Gelman A, Meng XL (eds) Applied bayesian modelling and causal inference from missing data perspectives. Wiley, New York
Ichino A, Mealli F, Nannicini T (2005) Temporary work agencies in Italy: a springboard toward permanent wmployment?. Giornale degli Economisti e Annali di Economia 64(1):1–27
Ichino A, Mealli F, Nannicini T (2008) From temporary help jobs to permanent employment; what can we learn from matching estimators and their sensitivity?. J Appl Econ 23:305–327
Imai K, Van Dyk DA (2004) Causal treatment with general treatment regimes: Generalizing the Propensity Score. J Am Stat Assoc 99(467):854–866
Imbens GW (2000) The role of the propensity score in estimating dose-response functions. Biometrika 87:706–710
Imbens GW (2003) Sensitivity to exogeneity assumptions in program evaluation. Am Econ Rev 93(2):126–132
Jeffreys H (ed) (1961) Theory of probability. 2nd edn. University Press, Oxford
Joffe MM, Rosenbaum PR (1999) Invited commentary: propensity scores. Am J Epidemiol 150:327–333
Kluve J, Schneider H, Uhlendorff A, Zhao Z (2007) Evaluating continuous training programs using the generalized propensity score. IZA DP No. 3255
Lechner M, Imbens G (2001) Identification and estimation of causal effects of multiple treatments under the conditional independence assumption. In: Lechner M, Pfeiffer F (eds) Econometric evaluations of active labor market policies in Europe. Physica, Heidelberg
Lu B, Zanutto E, Hornik R, Rosenbaum PR (2001) Matching with doses in an observational study of a media campaign against drug abuse. J Am Stat Assoc 96:1245–1253
Mealli F, Pagni R (2001) Analisi e valutazione delle politiche per le nuove imprese. Il caso della L.R. Toscana n.27/93. IRPET
Meini MC (2001) Politiche per l'occupazione a scala locale. Valutazione del ruolo degli interventi per lo start-up d'impresa. Provincia di Massa, Osservatorio Provinciale Mercato del Lavoro, MIMEO, IRPET
Mitnik O (2008) Intergenerational transmission of welfare dependency: the effects of length of exposure. Working paper, University of Miami
Newey W (1994) Kernel estimation of partial means and a general variance estimator. Econ Theory 10:233–253
Pellegrini G (2001) La struttura produttiva delle piccole e medie imprese italiane: il modello dei distretti. Banca Impresa Società 20(2):237–248
Pellegrini G, Carlucci C (2003) Gli effetti della legge 488/92: una valutazione dell'impatto occupazionale sulle imprese agevolate. Rivista italiana degli economisti 2:267–286
Rosenbaum PR (1987) The role of a second control group in an observational study. Stat Sci 2(3):292–306

Rosenbaum PR (2002) Observational studies, 2nd edn. Springer, NewYork
Rosenbaum PR, Rubin DB (1983a) The central role of the propensity score in observational studies for causal effects. Biometrika 70:41–55
Rosenbaum PR, Rubin DB (1983b) Assessing sensitivity to an unobserved binary covariate in an observational study with binary outcome. J R Stat Soc Ser B (Stat Methodol) 45(2):212–218
Rubin DB (1974) Estimating causal effects of treatments in randomized and nonrandomized studies. J Educ Psychol 66:688–701
Rubin DB (1978) Bayesian inference for causal effects: the role of randomization. Ann Stat 6(1):34–58
Rubin DB (1980) Bias reduction using Mahalanobis's metric matching. Biometrics 36:295–298
Rubin DB (1987) Multiple imputation for nonresponse in surveys. Wiley, New York
Rubin DB (1990) Comment: Neyman (1923) and causal inference in experiments and observational studies. Stat Sci 5:472–480
Rubin DB (2008) For objective causal inference, design trumps analysis. Ann Appl Stat 2(3):808–840